

2. IT Infrastructure Services & Cloud Computing

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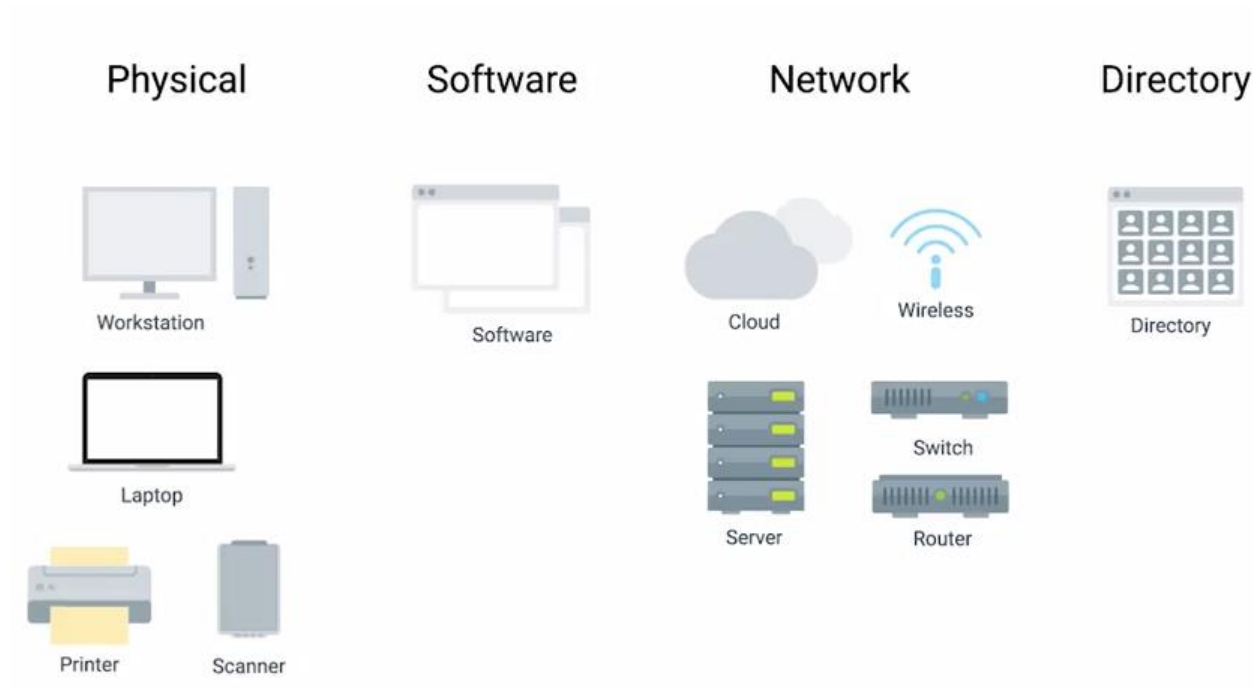
Learning Outcomes

- Explain IT infrastructure
- Compare cloud models
- Design cloud solutions
- Evaluate African cloud adoption
- Manage Linux cloud servers
- Understand cloud security

What is IT Infrastructure?

- Hardware, software, networks, storage, databases, applications, security and cloud resources.

IT infrastructure



University Infrastructure Example

- ERP, Moodle, Email, Wi-Fi, CCTV, Data Centers, Student Portals

Evolution of Infrastructure

- Traditional → Virtualized → Cloud Infrastructure

Evolution of Infrastructure

- Traditional Infrastructure (The Physical Era)
 - Traditional infrastructure relies on physical hardware.
 - **Bare Metal**: Applications run directly on physical servers.
 - **One Box, One Job**: one major application on one server. This prevents apps from crashing each other.
 - **Wasted Power**: Servers often sit idle. They use only a small part of their total power.
 - **Slow Changes**: Buying and setting up a new physical server can take weeks or months.

Evolution of Infrastructure

- Virtualized Infrastructure (The Software Era)
 - **Virtualization** uses software to split one physical server into multiple virtual servers.
 - **The Hypervisor**: This special software sits on top of the physical hardware. It lets you run many operating systems on the same machine.
 - **Virtual Machines (VMs)**: Each VM acts like a real computer with its own CPU, memory, and disk space.
 - **Higher Efficiency**: Companies can pack dozens of VMs onto one physical server. This saves money, space, and electricity.
 - **Faster Setup**: Creating a new virtual server takes minutes instead of weeks.

Evolution of Infrastructure

- **Cloud Infrastructure (The Modern Era)**
 - Cloud infrastructure takes virtualization and delivers it over the internet as a service.
 - **No Hardware to Own:** A cloud provider (like AWS, Google, or Microsoft) owns and manages the physical data centers.
 - **On-Demand Access:** Users can rent servers, storage, and databases instantly using a web browser or API.
 - **Pay-As-You-Go:** You only pay for the exact seconds or hours you use the tech.
 - **Instant Scaling:** The infrastructure can automatically grow or shrink based on how many people use your app

Cloud Computing Fundamentals

- On-demand self-service, broad network access, resource pooling, elasticity, measured service

Cloud Deployment Models

- Public, Private, Community, Hybrid

Cloud Deployment Models

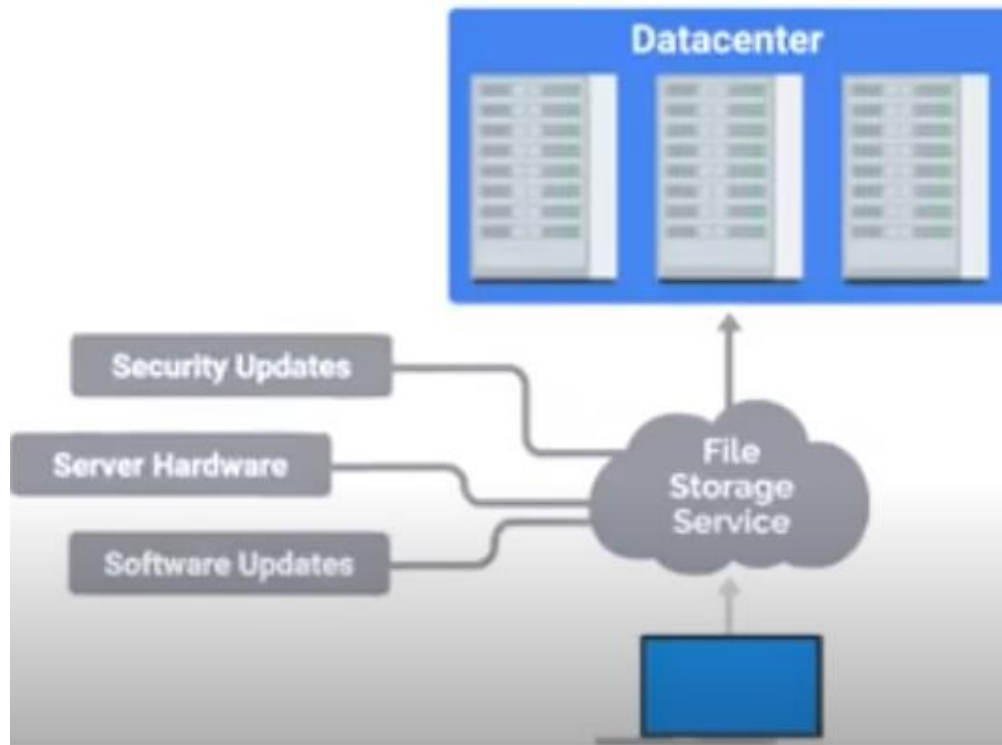
- **Public Cloud:** deployed by a provider to offer cloud services to the general public and organizations over the Internet.
- **Private Cloud:** set up for the sole use of a particular organization.
- **Community Cloud:** set up for the sole use by a group of organizations with common goals or requirements.
- **Hybrid Cloud:** A hybrid cloud is composed of two or more individual clouds, each of which can be private, community, or public clouds.

Public Cloud

- AWS, Microsoft Azure, Google Cloud; scalability and pay-as-you-go



An Amazon Data center (hundreds or even thousands of server machines)



Examples of services managed on the cloud

Private & Community Cloud

- Dedicated resources; examples include government and university shared infrastructure

Hybrid Cloud

- Mix of public and private resources; common in universities and enterprises

Cloud Service Models

- IaaS, PaaS, SaaS

Infrastructure as a Service

- Virtual machines, storage, networking; AWS EC2, Azure VM, Google Compute Engine

Platform as a Service

- Application deployment without server management; App Engine, Azure App Service

Software as a Service

- Google Workspace, Microsoft 365, Zoom, cloud ERP systems

Software as a service

SaaS: Software as a Service



Containers & Cloud Native

- Docker, Kubernetes, portability, scalability
- **Docker** and **Kubernetes** are tools that help run software in the cloud.
- **Portability** and **scalability** are the main benefits they provide.

Containers & Cloud Native

- **Docker**
 - Docker is a tool that wraps software into a neat package called a container.
 - This package holds the code and everything the code needs to run.
 - Because everything is inside the package, the software runs the same way on any computer.
- **Kubernetes**
 - Kubernetes is a tool that manages those Docker containers.
 - If you have hundreds of containers, it is hard to watch them all.
 - Kubernetes automatically starts them, stops them, and fixes them if they crash.

Cloud Networking

- VPCs, VPNs, Load Balancers, DNS, Firewalls

Cloud Storage

- Object, Block and File Storage

Linux in the Cloud

- Open source, secure, stable, automatable, efficient

Linux Distributions

- Ubuntu Server, Debian, RHEL, Rocky Linux

Linux Administration Skills

- SSH, SCP, systemctl, cron, package management

Cloud Security

- Shared responsibility, MFA, encryption, backups, logging

Kenya & Africa Cloud Adoption

- Banks, universities, fintech, government, regional data centers

African Cloud Providers

- Liquid Intelligent Technologies, Africa Data Centres, Raxio, KENET, Konza Technopolis

Emerging Trends

- AI Infrastructure, Edge Computing, Serverless, DevOps, Green IT

Discussion Questions

- Why Linux dominates cloud? Should African governments promote local cloud hosting?
- Differentiate between IaaS, PaaS, and SaaS.
- Explain the advantages of Linux in cloud environments
- What is a hybrid cloud?
- What is virtualization?
- Explain the shared responsibility model.

Assessment Activities

- A university wants:
 - Moodle
 - Student ERP
 - CCTV storage
 - Email services
- Design a university cloud architecture and justify deployment choices

Assessment Activities

- Investigate one organization in Kenya that has adopted cloud computing and discuss:
 - Reasons for adoption
 - Benefits achieved
 - Challenges encountered

Further Learning Resources

- AWS Skill Builder, Microsoft Learn, Google Cloud Skills Boost, Linux Journey